Backtranslation as a Remedy for Low-Resource Language Pairs in Machine Translation — Ndyuka as a Case Study

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1 Introduction

The broad field of Language Technology (LT) has, in recent years, seen the introduction of new developments across various of its subfields that all combat the data hungriness of modern, large neural networks. For example, the *pretraining-finetuning* paradigm popularized in the field of LLMs has opened up ways to train large models for new tasks with less data and resources than before, Parameter Efficient Finetuning (PEFT) methods such as LoRA (Hu et al., 2022) have made continued training of existing models much more computationally and data efficient, and the introduction of adapters in large ASR models such as in the work of Pratap et al. (2024) have made it possible to create new ASR solutions for languages that didn't have such solutions before due to lack of data. This latter development is an example of a broader shift within the LT research field towards *low-resource languages*, languages for which exist little data to create ML-based solutions with: as it has become more feasible to create LT solutions for languages with little data, and as large, high-resource languages now have relatively robust LT solutions, low-resource languages are increasingly researched. This is also true in the field of Machine Translation (MT), where research on the automatic translation from, to, and between low-resource languages is becoming increasingly popular. This paper is an example of such research, as we discuss the development of MT solutions for the Ndyuka language. Specifically, in this work, we ask ourselves the following question:

> *What is the change in performance for our Ndyuka to Chinese/German machine translation systems when backtranslated sentence pairs are added to the training data?*

2 The Ndyuka Language

Ndyuka, also known as Eastern Maroon Creole or Aukan, is a language spoken by the Ndyuka people of Suriname and French Guiana. The Ndyuka people are descendants of Africans who were enslaved by the Dutch and brought to Suriname to work on plantations, and then were able to escape slavery to live in the Surinamese interior in communities. Due to its speaker's history, the Ndyuka language is a creole language with roots in African languages which has been influenced significantly by Western European languages. Specifically, Ndyuka grammar and phonemes can be traced back to African languages, while Ndyuka vocabulary is mostly based on English, Portuguese, and Dutch (Borges, 2014). Although a syllabary for Ndyuka exists, it is most often used using the Latin script, and this is reflected in digitalized Ndyuka resources. The use of the Latin script as well as the heavy influence of English, Portuguese, and Dutch on the vocabulary contribute to Ndyuka using mostly characters that are also used in the Western European languages mentioned: the Ndyuka orthography uses the 26 letters of the Latin alphabet with some omissions such as *R*, *C*, and *Q*, and uses a few digraphs that end in *Y* for a few phonemes, such as *TY* and *DY* (such as in *Ndyuka*). Furthermore, long vowels in Ndyuka are written by doubling letters and while tone is used in spoken Ndyuka, it is rarely written in Ndyuka texts.

3 Related Work

Our work is an extension of the work by Robinson et al. (2024). In their work, Robinson et al. (2024) create a dataset of aligned sentences between a few high-resource languages, such as English, French, German, and Chinese, and a large variety of creole languages, focusing on creoles of the Caribbean and surrounding areas. They include data of 41 creole languages total in their dataset, of which 21 creoles did not yet have aligned sentence data before. The publicly available part of their aligned sentence dataset was also published on the HuggingFace platform.[1]

Furthermore, Robinson et al. (2024) also published several Neural Machine Translation (NMT) models for many of the language pairs covered by their creole dataset. These are versions of the NMT model mBART50 by Tang et al. (2021) that were finetuned further on the aligned sentences of the creole dataset. mBART50 itself is a Transformer-based (Vaswani et al., 2017) model that was pretrained on a denoising task and then finetuned for the Machine Translation task.

Relevant to our work is the concept of backtranslation. **Backtranslation**, within the context of Machine Translation, is the process of translating monolingual text data in the target language to the source language to create new aligned sentences. These backtranslated sentences can then be used to train an MT model from the source language to the target language. Backtranslation can be especially fruitful if the target language is a high-resource language while the source language is a low-resource language, such as in the case of many language pairs of the creole dataset by Robinson et al. (2024), where creoles are translated into high-resource languages. Edunov et al. (2018) describe how backtranslation can be performed by NMT models and what the efficacy of such methods is.

4 Methods

Our work focuses on if and how supplementing a low-resource language like Ndyuka with additional training data using backtranslation can improve translation results. Specifically, we focus on using the data of the creole dataset by Robinson et al. (2024) and on translations into German and Chinese. We chose these two target languages as every author understands at least one of these languages to a degree where automatic translations could be personally assessed on perceived quality, and because we wanted to forego translation into English since that is already a relatively widely-covered topic. We found Haitian and Ndyuka as the two source languages that fulfilled our constraints. We chose Ndyuka as our source language because we perceived it as lower-resource than Haitian in the creole dataset: while about 280,000 aligned sentences are available for Haitian $\leftrightarrow$ German and 50,000 for Haitian $\leftrightarrow$ Chinese, only about 2,000 sentences are available for both Ndyuka $\leftrightarrow$ German and Ndyuka $\leftrightarrow$ Chinese. In our eyes, this meant Ndyuka could benefit more from backtranslations.

For our research, we focus on using both Statistical Machine Translation (SMT) approaches as well as an NMT approach. We build two SMT models from scratch using the Ndyuka $\leftrightarrow$ German/Chinese data with additional backtranslations. Our reasoning behind going with an SMT approach is because the creole dataset has only $\sim$2000 training pairs for Ndyuka $\leftrightarrow$ German/Chinese, and as Koehn and Knowles (2017) demonstrate, NMT models really struggle to perform with such limited data. For this reason, we do not build an NMT model from the ground up. Instead, we apply two different approaches:

(1) Take creole NMT models trained by Robinson et al. (2024) and continue their finetuning on backtranslated data. This gives us the existing creole NMT models as a baseline to compare our models to, as well as the opportunity to apply potential backtranslation benefits to an NMT model.

(2) Finetune a high-resource NMT model to adapt it to Ndjuka, based on the experiments presented in Neubig and Hu (2018).

5 Experiments

5.1 SMT: Ndyuka & Chinese

5.1.1 Dataset

The Ndyuka-Chinese dataset includes 2220 entries in total, also mostly related to Christianity. Using the Bible is an effective way to make parallel data because as long as one has a sentence in the Bible, it is always possible

[1] https://huggingface.co/datasets/jhu-clsp/kreyol-mt

to search the corresponding entry in another language.[2]

Source	2 Corinthians 11:7, King James Bible
Ndyuka	Ma u si enke na ogii taki, mi be saka miseefi du son wooko enke ibii taawan de? We, na fu be soi fa u mu libi fu taampu hei gi Masaa Gadu de. Ne mi gi u Masaa Gadu Bun Nyunsu sondee fu meke u pai mi fu dati.
Chinese	我因为白白传神的福音给你们、就自居卑微、叫你们高升、这算是我犯罪么
English	Have I committed an offence in abasing myself that ye might be exalted, because I have preached to you the gospel of God freely?

However, due to the different styles for different translations, sometimes the translations themselves do not match. In the example, the Chinese version uses 'feet' like English, but in Ndyuka, instead of using 'futu', they use 'sikin', which means 'body'.[3] The Chinese version of the Bible contains many remarks, always inside parentheses, which are basically paraphrases about specific expressions. As they cannot and should not be aligned, they should be stripped from the data.

Source	Hebrews 12:13, King James Bible
Ndyuka	Da waka bun meke swakiwan di seefi de fu dyakata komoto sa teke taanga sikin a yu. Da den sa kon taanga baka, fu biibi go doo.
Chinese	也要为自己的脚把道路修直了、使瘸子不至歪脚、反得痊愈。〔歪脚或作差路
English	And make straight paths for your feet, lest that which is lame be turned out of the way; but let it rather be healed.

Another main problem is punctuation[4]. Whenever there is punctuation other than a full stop or a comma, there will always be something wrong, either missing the closed one or using non-standard punctuation or both. The use of full stop is messy in the data. Usually, the full stop is missed, so the sentence ends without punctuation. The correct full stop for Chinese language is a circle called ideographic full stop '。', instead of a solid dot. However, in some cases, the fullwidth full stop '．' is used. As they are all presented by different Unicode points, it should cause some problem in training.

Source	Hebrews 7:17 King James Bible
Ndyuka	Bika Masaa Gadu Buku sikiifi taki: "Yu o de a moo hei apaiti begiman gi taawan a mi Masaa Gadu ya fesi, fu tego, Dati sa waka na a fasi fu Melsisedek."
Chinese	因为有给他作见证的说、『你是照著麦基洗德的等次永远为祭司。
English	For he testifieth, Thou art a priest for ever after the order of Melchisedec.

5.1.2 Extra data

For statistical models, more data are required to train an n-gram model. As all data provided for this task are from the Bible, the extra corpus chosen was also a Bible corpus.[5] As mentioned before, the language style used in the Bible looks different with the language people using today. The corpus provides both the Chinese and the German version. For the Chinese one, it contains more than 30 thousand lines of sentences, which should be large enough for an n-gram model. However, the data is not perfect. The main problem is that it uses Traditional Chinese, so a tool[6] was adopted to convert the characters to Simplified Chinese. Also, the data is in XML format, which means that a parser was created to extract sentences.

[2] All English references in examples come from https://cnbible.com/.

[3] https://suriname-languages.sil.org/Aukan/English/AukanEngLLIndex.html

[4] Strictly all sentences are wrong. In modern Chinese, the fullwidth comma '，' should be used rather than the ideographic comma '、'. However, as this punctuation appears for all sentences, the sentences can be treated as from an old text, then it doesn't matter anymore.

[5] https://github.com/christos-c/bible-corpus

[6] https://pypi.org/project/langconv/

5.1.3 Models

The system for training the SMT models is MOSES, which can be used for any language pair. The model itself is simple to use, because for training, only a parallel corpus is necessary, which we have with the Ndyuka ↔ Chinese dataset. However, to use this system, an n-gram model for the target language is required. The data prepared for the n-gram model was from another Bible corpus. With more than 30 thousand sentences, it should be large and relative enough to build a decent n-gram model. The data was cleaned before training. As mentioned before, the punctuation of the Chinese data is in poor situation, which might have a impact on the model. For Ndyuka, due to the lack of existing tools, the processing was done by treating the language as English, as that is the language that had the most influence on Ndyuka's vocabulary.

Besides, for a traditional way of Chinese language processing, the segmentation is always troublesome. Different tools will produce different segmentation, and different segmentation will cause different results. In the experiment, two popular tools, one called *jieba*,[7] and another, called *THULAC* (Li and Sun, 2009), were used to test the influence of segmentation. The final decision was to adopt *jieba*, which seemed to lead to a better model in this case, although *THULAC* claims to have better performance.

For the back-translation part, all steps should be the same, except the part of introducing another n-gram model for the source language, as the result of the first run needs to be translated back. However, due to some technical issues, the first n-gram model was built with KenLM,[8] compiled on our own machine, while the second one was from SRILM, which was accessible for everyone in the project. For an n-gram model, different tools might produce their models with different speed, but the output themselves should be the same. However, it is still necessary to mention the incident here. Also, with the lack of confidence on the processing of Chinese language of the model, another experiment was performed with the provided data extended with data from the extra Bible corpus. While translating the extra data from Chinese to Ndyuka, English was adopted as the middle language. The Chinese-English part was done by the mBART-50 model, which has already performed well for this language pair. Then the result was concatenated into the provided training data. In this case, the training data was able to get expanded to more than 32 thousand lines.

[7] https://github.com/fxsjy/jieba?tab=readme-ov-file#jieba-1

[8] https://github.com/kpu/kenlm

5.2 SMT: Ndyuka & German

5.2.1 Dataset

The Ndyuka-German dataset contains 1860 sentence pairs – 1640 / 76 / 146 in the train / validation / test sets respectively. Like the Chinese data, they are made up exclusively of bible verses.

The German verses correspond to the second revision of the Lutheran bible which was published in 1912 (Kähler, 2024). For a native speaker the verses are thus mostly understandable, but they also regularly contain outdated terms and stilted language – most likely in an effort to not stray to far from the original verses. Examples are words like *alsobald*, *Scheltwort*, *stäupen*, *ward*, which can all be categorised as either old-fashioned (Dudenredaktion, a,b), or even outdated (Dudenredaktion, c,d).

5.2.2 Extra Data

As mentioned in section 5.1.2, the additional monolingual German data for the language model is again taken from the Bible corpus by Christodouloupoulos and Steedman (2015). Using Bible verses as the basis of the LM helped to ensure that more archaic words – as the ones described in section 5.2.1 – were part of the LM's vocabulary.

5.2.3 Models

The same SMT pipeline using MOSES was applied (see 5.1.3).

During preprocessing punctuation was normalized, and then the data was tokenized and lowercased. While the tokenizer can handle German, it was again set to English for Ndyuka.

For the baseline LM a ready-to-use German

LM was used.[9] After SRILM was available, we built a German *'Bible-LM'*[10] from the aforementioned Bible corpus, thus adapting to the domain and expecting better results.

The Ndjuka LM for the back-translation was trained with SRILM on 11.8k Ndjuka sentences taken from the significantly bigger English ↔ Ndjuka data set.[11]

5.3 NMT: Ndyuka & German

In addition to our SMT solutions, we continued the finetuning of the creole models provided by Robinson et al. (2024) on a backtranslated dataset of Ndyuka ↔ German examples. We created this backtranslation dataset using the Ndyuka ↔ English aligned sentences of the creole dataset, and translating the English sentences into German using the creole mBART50 model finetuned on publicly available creole data. Since the Ndyuka ↔ English subset consists of about 11,800 aligned sentences, this created a dataset of about 11,800 backtranslated Ndyuka ↔ German aligned sentences. This dataset has been made publicly available through HuggingFace[12].

Two versions of the creole mBART50 NMT models were finetuned on this backtranslated dataset: one that was finetuned on publicly available data only ("pubtrain") and one that was finetuned on both public and private data ("full"). Both model versions were finetuned for 5 epochs on the backtranslation dataset using LoRA (Hu et al., 2022). We specifically used LoRA to obtain updated weights for the attention modules of the mBART model, which is only a small part of the entire model. This resulted in training times of about 10 to 15 minutes on an NVIDIA RTX A6000 for both models. Additionally, the model trained on publicly available data only was also trained for 20 epochs on the backtranslated dataset, which took almost an hour to complete on the A6000 GPU.

[9] https://sourceforge.net/projects/cmusphinx/files/Acoustic%20and%20Language%20Models/German/

[10] Trigram model with Kneser-Ney smoothing and -*unk* option to replace out-of-vocabulary words

[11] It should be noted that this data actually also contains non-Bible sentence pairs

[12] https://huggingface.co/datasets/daanbrugmans/kreyol-mt-backtranslations

5.4 NMT: Ndyuka & German – OPUS

Following one of the ideas in Neubig and Hu (2018), the idea in this second NMT approach is to start with a trained high-resource MT-model that fulfills two criteria; (1) Its source language should be related to the source language of the model we are aiming toward, i.e. Ndjuka. (2) Its target language should be our target language, i.e German. As Ndjuka is a Creole heavily drawing on English vocabulary (Huttar and Huttar, 2003), the *opus-mt-en-de* model[13] (Tiedemann and Thottingal, 2020) fits the bill.

After a different preprocessing with the SentencePiece tokenizer (Kudo and Richardson, 2018), we finetune that model with the Ndjuka-German training pairs. The similarity of English and Ndjuka ideally allows for some knowledge to transfer and lead to decent results, even with our severely limited data set.[14]

6 Results

6.1 SMT: Ndyuka & Chinese

Models	Baseline	Back-translation	Middle language
Multi-BLEU	1.98	2.03	2.45

Table 1: The result for the Chinese-Ndyuka models

The results are presented in Table 1. The baseline, which was trained with the provided data only with tuning achieved multi-blue score 1.98, while the back-translation model 2.03 and the one with middle language 2.45. The MOSES system itself can be easily blamed, arguing that the model is not good at handling Ndyuka and Chinese language. However, this theory was examined while doing the back-translation. The Multi-BLEU score of the output of the Chinese-Ndyuka model reached 9.47. The result confirms the capacity of not only Chinese but also any new language for MOSES. Ignoring the size and the quality of data, the only option then is segmentation.

By checking the phrase table, it is easy to discover that the alignments are messy,

[13] https://huggingface.co/Helsinki-NLP/opus-mt-en-de

[14] This approach corresponds to the *cold-start Bi−→Sing* adaptation in Neubig and Hu (2018)

especially for short phrases. For example, the phrase 'be de'[15] got matched to *is/are*, *was/were*, *have*, *already had*, *also*, *when*, *alive until*, and even *empty place*.

Segmentation definitely caused a huge impact on the result, but it can also be the difference of language systems. In the Chinese language, there is no concept of tense, so how can it match the word *be* for the past tense? It can either find some similar words, like the particles 了 indicating the perfect aspect, or some words around the corresponding location, making the result weird.

6.2 SMT: Ndyuka & German

As presented in Table 2, the results closely resemble those of our first language pair. The back-translation only has a very minor effect, while the switch to the Bible-LM changes the Multi-BLEU score more significantly.

Expanding the back-translation to the full 31.1k additional sentences, does not help to improve the model further. Drowning out the original data with 20x as many suboptimal back-translated pairs actually significantly worsens performance instead.

Tuning the SMT with MERT also did not improve the BLEU scores, which can probably be attested to the limited validation set of only 76 sentence pairs.

The Bible-LM helps with the domain-specific, and often old-fashioned words alluded to in Section 5.2.1. For example, we only find *alsobald* or *daß* (an outdated spelling of the common German conjunction *dass*) in the output of the Bible-LM supported models, and not in the baseline models.

Even the translations of the best model are obviously still quite poor. The generations are too long – 70% more tokens than in the reference – and most matches contributing to BLEU are unigrams (25%), while bi-, tri-, and fourgram precision is low with 5.3, 1.8, and 0.8% respectively.

Skimming the German translations also reveals several instances of repeated words, for example (English Bible verse as reference):

Ndyuka	ne **wanten wanten** a sikiman kon bun baka...
German	und **alsbald alsobald** der sikiman heil wieder...
English	**At once** the man was cured...

Here, the word *wanten* is reduplicated to intensify its meaning – through augmentative reduplication (Huttar and Huttar, 1992) – from *soon* to *at once*, which roughly corresponds to the German *als(o)bald*. This reduplication should not be present in a correct German translation, but since the phrase table contains *wanten* almost exclusively with *als(o)bald*, and never with *bald* (= *soon*), this error follows.

Ndyuka	u de **sete sete** , fu kaasi ibii taanga...
German	ist **nun** ihr **nun** , zuwider tun in allen...
English	And we will **be ready** to punish every...

In this second example, *sete* is a word that usually marks the ingressive aspect of action verbs (Huttar and Huttar, 2003), as in the action is beginning or someone is getting ready to do it. In translation this is usually expressed via the German equivalent to *now*[16] – *nun* – which the phrase table confirms; *sete* is paired with *nun* 59 times and only 7 times with another token. But in this case the reduplication of *sete* again changes the meaning, where *sete sete* then represents the state of *being ready*, which the German (nonsensical) translation does not represent.

6.3 NMT: Ndyuka & German

Table 3 shows the results of the NMT experiments expressed in the BLEU (Papineni et al., 2001) and chrF (Popović, 2015) metrics. This includes a comparison between the models provided by Robinson et al. (2024) ("Out-of-the-Box") and those finetuned on the backtranslated dataset ("Finetuned"). Metrics were calculated using the test split of the Ndyuka ↔ German dataset provided by Robinson et al. (2024).

Although the "Full" model has a slight decrease in performance after being finetuned on the backtranslation dataset, the creole

[15] *be* indicates past tense, and *de* means *be* in English.

[16] When an action is beginning it is happening *now*

Train Data / LM	Base data	+1.64k back-translated pairs	+31.1k back-translated pairs
Baseline LM	1.11	1.41	0.47
Bible LM	3.10 / 2.94 (MERT)	**3.68** / 3.11 (MERT)	1.23

Table 2: Multi-BLEU scores for the German-Ndyuka models on the test set

	Out-of-the-Box		**Finetuned**		
	Full	*Pubtrain*	*Full*	*Pubtrain (5)*	*Pubtrain (20)*
BLEU	74.5	16.3	71.3	24.7	59.7
chrF	81.6	43.1	80,5	49.2	74.5

Table 3: BLEU and chrF scores for the NMT models. “Out-of-the-Box“ indicates models not yet finetuned on the backtranslated dataset, “Finetuned“ models are finetuned on the backtranslations. “Full“ indicates models finetuned on public and private creole data, “pubtrain“ on just public. Numbers in parentheses indicate number of epochs trained.

mBART model trained on only the data available in the dataset published by Robinson et al. (2024) (“Pubtrain“) does improve by 8.4 BLEU points and 6.1 chrF points after being finetuned using LoRA for only 5 epochs. This means that finetuning using LoRA on the backtranslation dataset for only 5 epochs on a dataset of 11,800 sentence pairs improved the NMT model’s BLEU score by 51.5%. This improvement continues when the model is trained for more epochs, as the 20 epoch Pubtrain model sees a BLEU score improvement of about 266% over the original Pubtrain model, and a 73% increase in chrF score over the original Pubtrain model, with metric performance closing in on the models trained on both public and private data.

However, it should be noted that the metrics shown here, especially those for the Full models and the Pubtrain model trained for 20 epochs, are surprisingly high. We suspect that this is due to the homogeneity of the data: most of the data for both the original Ndyuka ↔ German dataset and Ndyuka ↔ English dataset consist of phrases from the Bible, and thus so does the backtranslation dataset. This means that most of the dataset falls within one specific domain, and this topic homogeneity combined with the limited data likely makes many sentences easy to predict by a model after a set amount of training epochs. To get a better insight into the models’ performances, metric calculation would preferably be performed on manually translated sentence pairs about a different topic.

6.4 NMT: Ndyuka & German – OPUS

While this second approach impressed with its simplicity, it seemed to not be sufficient in the context of the limited data we were working with. While the loss was lowering during training, the BLEU score on the evaluation score was fluctuating and never managed to become larger than 2.0 – across different runs, hyperparameter settings, and making use of early stopping during tuning. The model likely successfully memorized the small training set, but was never able to considerably generalise beyond it, and capture more of the Ndjuka language.

Introducing German back-translated data from the German -> Ndyuka SMT did also not improve the model. Yet, the final BLEU score of **2.33** is notably still better than our baseline SMT model without the Bible-LM support.

7 Discussion and Suggestions

For both SMT and NMT, with such small amounts of data, results are not expected to be good. For the Chinese and German SMT models, the back-translation itself did not contribute too much in this case, and at least from the results, the improvement is not significant, e.g. compared to the clearer impact of a general vs the domain specific Bible-LM. It is by adding more data that the models are able to gain a little better performance.

On the other end of the spectrum, the mBART NMT model showed surprisingly high metric scores, likely due to the small size and topic homogeneity of the data, since both train and test data are mostly Bible phrases. As a suggestion for the future, we would want to test our models on data of different topics, as we think that would give a better representation of the models' true, general performance.

The performance of the second NMT approach was less impressive. One of Neubig and Hu's (2018) models, which adapted a Turkish -> English NMT model to Azerbaijani -> English, had a BLEU score of 8.7 on only about 6k train sentences. We assume that 3 factors limited the performance in our case (1) We have even less training data (1.6k) (2) Our target sentences contain domain-specific, sometimes outdated, language which means we introduce new things on both side of the model, not just on the input (3) Azerbaijani & Turkish are more similar than English & Ndjuka, limiting our model's ability to transfer knowledge.

However, does it mean by adding a middle language, the problem of low-resource can be solved? The answer should be negative. On paper, the number does get improved, but remember how much data was concatenated, plus the time spent for running a transformer model for the middle language. With such a low-resource situation, the basic idea is usually to find a method to produce more artificial data for training, but no matter in which way, the artificial data is usually in poor quality. Then, even with 31 thousand more sentences, and a whole day for training, the improvement a model can make is still limited.

Also, does it mean that the back-translation is not suitable for a low-resource language? The answer is still unclear, as it shows some potentials on the NMT model. Surely a plain back-translation might not be able to produce obvious improvements for the translation, but, in the future, with more and more optimization methods for finetuning a low-resource model with back-translation being investigated, it could be a fast and easy way to get a decent model for low-resource translation.

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Dudenredaktion. a. alsobald rechtschreibung, bedeutung, definition, herkunft.

Dudenredaktion. b. Scheltwort rechtschreibung, bedeutung, definition, herkunft.

Dudenredaktion. c. stäupen rechtschreibung, bedeutung, definition, herkunft.

Dudenredaktion. d. ward rechtschreibung, bedeutung, definition, herkunft.

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